Discrete Drainage Dynamics III:
A Floquet Two-Step Spinor Sector and Lorentz-Like Kinematic Time Dilation

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Abstract

Discrete Drainage Dynamics (DDD) is an analogue-substrate programme. It does not contest special relativity or Lorentz invariance: those are empirical anchors that the analogue must reproduce in the appropriate continuum limit. We ask the narrower, exploratory question of whether the kinematic time-dilation factor of special relativity can arise operationally from a discrete Floquet spinor rule inside the toy substrate.

Papers I and II [1, 2] treated the substrate as a scalar reserve field $R_i \geq 0$ with first-order time dynamics. That sector cannot support propagating localised excitations: the rule is diffusion-like and any initial bump spreads. Here we extend the substrate with a two-component spinor $\psi_i \in \mathbb{C}^2$ at each node, and a Floquet two-step time evolution that alternates a matter-hopping update A and a gauge-twist update B with period $T = \tau_A + \tau_B$. Within the class of local stateful-link rules satisfying four operational propagation axioms (F1)–(F4), this two-step structure is selected as the minimal two-component Pauli/Floquet representative; closing the construction further requires a topological hypothesis (F-twist) that fixes the diagonal-twist normalisation $\lambda = 1/(2\pi)$. The Floquet quasi-energy then exhibits a band gap of magnitude $m = \lambda/T$ at the lattice point $(\pi, 0)$, around which the dispersion is approximately Lorentz-like, $\varepsilon^2 \approx m^2 + (c_{\text{eff}}\, p)^2$ with $c_{\text{eff}} = 1/T$ and explicit quartic lattice corrections $O(\delta^4)$ in the momentum offset. Localised wavepackets near this gap point propagate at the predicted group velocity, and we measure operationally the kinematic clock-rate $m/\gamma = m^2/E$ at moving packets via a Galilean-shifted, dealiased k-space projection. The measured proper-time rate matches the prediction m/γ to better than 1% for $\gamma < 2$ (median deviation $\sim 0.3\%$ over $\gamma \in [1, 2]$) and within $\sim 10\%$ for $\gamma < 4$, with monotonic deviations beyond. The result is the kinematic counterpart of the bandwidth identification of Paper II: the moving packet's effective clock runs at m/γ in lab time, measured from substrate observables without invoking Lorentz transformation laws. We do *not* derive the Lorentz group, the spatial part of the metric, the Einstein equations, or the spinor sector itself; their derivation is deferred to Papers IV–VII. The (F-twist) normalisation is the principal open problem: the topological hypothesis fixing $\lambda = 1/(2\pi)$ closes the construction but remains microscopically unexplained.

What this paper makes mechanically legible. Special relativity postulates the existence of a finite invariant speed c as an empirical input; SR/GR has no structural explanation for *why* a propagation ceiling exists, and encodes kinematic time dilation as a metric transformation law. *This paper* forces the existence of a propagation ceiling $c_{\text{eff}} = 1/T$ from finite per-tick lattice bandwidth (locality (L1) + (L5)), and reads the kinematic factor m/γ as the operational comoving phase rate of a Floquet wavepacket — no Lorentz transformation invoked. **Leverage:** the *existence* of c becomes a substrate consequence rather than an ad-hoc postulate, and the SR transformation law is replaced by an operational phase-rate measurement on a discrete medium.

1 Main claims and scope

Programme positioning and analogue framing. Discrete Drainage Dynamics is an analogue-substrate programme; it does not contest special relativity or Lorentz invariance. Those are the empirical anchors the analogue must reproduce. The aim of this paper is the narrower question of whether the *form* of the kinematic time-dilation factor can emerge operationally from a discrete Floquet spinor rule. The result is internal to the analogue, conditional on the axioms (F1)–(F4) and the topological hypothesis (F-twist) introduced in Sec. 3, and holds only in the continuum limit and for the specific spinor sector adopted. Any deviation of the analogue from special relativity at the level of measured tests is a falsifier of the analogue, not of SR.

This paper introduces a structurally new sector of the substrate beyond the scalar reserve of Papers I and II, and uses it to recover a Lorentz-like kinematic clock-rate relation $\omega_{\text{comov}} = m/\gamma$ at moving wavepackets.

1. **Spinor sector.** The substrate carries, in addition to the scalar reserve R_i of Paper I, a two-component $\mathbb{C}^2$ spinor field $\psi_i = (a_i, b_i)^\top$ at each node, evolving under a Floquet two-step rule. The spinor sector is decoupled from the scalar reserve in this paper; coupling is deferred to Paper VII.

2. **Lorentz-like dispersion.** The Floquet quasi-energy of the spinor sector exhibits a band gap of magnitude $m = \lambda/T$ at the lattice momentum $(\pi, 0)$, with $\lambda = 1/(2\pi)$ from (F-twist) and $T = \tau_A + \tau_B$. Near this gap point, the dispersion is approximately

$$E^2 \;\approx\; m^2 + c_{\text{eff}}^2\, p^2 + O(\delta^4), \qquad c_{\text{eff}} = 1/T, \tag{1}$$

where $p = k - k_0$ is the momentum offset from the gap and the $O(\delta^4)$ correction is computed analytically in Sec. 7. Excitations near the gap are massive, relativistic, and propagate at the group velocity $v_g = \partial E/\partial p$.

3. **Operational kinematic clock identification.** For a localised wavepacket centred at momentum p with energy $E(p)$ and group velocity $v_g(p)$, the dealiased Galilean-shifted projection

$$A_{\text{comov}}(t) \;=\; \langle k_0|\psi(t)\rangle \cdot \exp(-i\, p \cdot v_g\, t)$$

has a phase rate equal to $E - p \cdot v_g = m^2/E$ in the continuum Lorentz limit. This is the kinematic counterpart of the bandwidth identification of Paper II: the operational *proper-time rate* of the moving packet is m/γ.

4. **Numerical verification.** The measured proper-time rate matches m/γ to better than 1% for $\gamma < 2$ (with relative deviations in $[0.01\%, 1.3\%]$, median $\sim 0.3\%$) and remains within $\sim 10\%$ up to $\gamma = 4$. Beyond $\gamma \sim 5$, lattice corrections to Lorentz become significant; we quantify them analytically (quartic in the momentum offset, Sec. 7) as a structural prediction of the discrete substrate.

Limitations and scope.

We do not derive the Lorentz group, special relativity, or covariance under Lorentz transformations of any field. Within the candidate spinor sector adopted in Sec. 3, we recover the operational relation $\omega_{\text{comov}} = m/\gamma$ in the continuum limit of the lattice rule. We do not derive the spatial part of the metric, the photon sector, or the Einstein field equations. The derivation in Sec. 3.1 establishes that the Floquet two-step structure with a 2-component Pauli spinor is the minimal representative within the class of local stateful-link substrates satisfying four operational axioms (F1)–(F4). However, the diagonal-twist normalisation $\lambda = 1/(2\pi)$ is not derived from those axioms: it is adopted as a *topological hypothesis* (F-twist) on the unit-flux quantisation of the diagonal-twist sector, and constitutes the principal open problem of the construction. We do not address the coupling between the spinor and the scalar sector: in this paper they are independent. The coupling and any related conservation laws are deferred to Paper VII; the schematic combined gravitational–kinematic factor of Sec. 8 is a *conjecture* for that paper, not a verified result of the present one.

Motivation and significance

Special relativity is one of the most mysterious empirical facts of physics for everyday intuition: clocks slow with motion, lengths contract, mass is convertible into energy via $E = mc^2$, and the speed of light is the same for every inertial observer. In standard physics, Lorentz invariance is an *input postulate*: the Minkowski metric and the universal c are taken as given, and the kinematic consequences (time dilation, length contraction) follow mathematically. The framework is descriptive, not derivative.

This paper proposes a different reading of the kinematic phenomena. The substrate of Paper I is supplemented with a Floquet two-step spinor sector — the same substrate, with two interleaved tick phases (A and B) and a two-component spinor ψ on each node. This sector supports propagating localised excitations whose band-structure is that of a 3D Weyl semimetal. Near the Weyl points, the dispersion is linear and isotropic: $\omega(\mathbf{k}) = v_g\,|\mathbf{k}|$. An operational clock measurement on these excitations — comoving with the wavepacket — yields a *kinematic* clock-rate dependence that matches the Lorentz form $\sqrt{1 - v^2/c^2}$ at the percent level for $v < 0.5\,c$.

This is not a derivation of full Lorentz invariance: the substrate breaks exact Lorentz at high momenta where lattice anisotropy dominates, and the spinor sector is a structural extension beyond the scalar reserve of Paper I. But it shifts the kinematic mystery: in DDD, time dilation is not a consequence of postulated Lorentz invariance, it is an operational consequence of how propagating excitations are clocked on a discrete substrate. The deeper question — why the substrate admits the Floquet 2-step structure — is the principal open question of this paper.

Which mystery does this paper address? Per SERIES_FEEDBACK §0.bis, this paper targets:

- **Mystery #1 (the speed-of-light ceiling).** The Floquet two-step rule has a finite group-velocity ceiling $c_{\text{eff}} = 1/T$ set by the substrate's tick period. We identify this ceiling as the analogue counterpart of the empirical c. The paper does not derive the CODATA value of c in physical units; it derives the existence of the ceiling from finite per-tick bandwidth.
- **Mystery #5 (operational origin of inertia and the kinematic clock-rate).** SR/GR encode both through metric postulates. We ask whether a kinematic time-dilation factor of Lorentz-like form (m/γ) arises operationally from a discrete Floquet spinor rule on a lattice. The answer offered here: under axioms (F1)–(F4) and the topological hypothesis (F-twist), the analogue's comoving phase rate matches the form of m/γ in the continuum limit near a band gap. The microscopic origin of (F-twist) (and hence the value of $\lambda = 1/(2\pi)$) is open and is the principal *why*-question this paper does *not* resolve.

Distinctive contribution and ontological reading. Special relativity postulates Lorentz invariance and the existence of a finite invariant speed c. Within SR/GR, there is no structural explanation of *why* c exists, or why it has the role of a propagation limit; Lorentz transformations are taken as postulates, not derived. Likewise, kinematic time dilation $\sqrt{1 - v^2/c^2}$ follows from the metric's indefinite signature, not from a mechanism. These are the distinctive points where the present paper's analogue contributes:

1. **Structural origin of the speed-of-light ceiling.** SR/GR introduces c as an empirical postulate; there is no mechanical answer in the standard formalism to "why does a

finite propagation ceiling exist at all?". In the analogue, the discrete tick τ and the per-link bandwidth α together force the existence of a bounded group velocity $c_{\text{eff}} = 1/T$ *by construction*: information cannot cross more than one lattice link per tick, hence cannot propagate faster than c_{eff}. **The existence of a finite speed-ceiling is therefore not postulated in DDD; it is a structural consequence of locality (L1) and finite per-tick bandwidth (L5).** Identification with the empirical CODATA c remains a calibration; the *existence* of the ceiling does not.

2. **Operational origin of the kinematic time-dilation factor.** In SR, m/γ is a transformation law: an inertial observer infers it via the Lorentz boost. Here, m/γ is the *operationally measurable* comoving phase rate of a moving wavepacket, $\omega_{\text{comov}} = E - p\,v_g = m^2/E$. No Lorentz transformation is invoked at any stage; the analogue produces the same factor as a substrate-level rate. **This displaces a transformation-law fact into a phase-rate measurement on a discrete medium.**

3. **Mechanical reading of inertia.** In GR/SR, inertia is encoded geometrically (the metric pairs energy and momentum). In the analogue, a faster-moving wavepacket has its lab-frame energy split between rest mass and kinetic momentum, and its comoving phase rate is correspondingly suppressed. The bandwidth interpretation reads "moving clocks tick slow" as "moving packets have less bandwidth available for their internal phase, the rest of which is spent on translating the centroid".

What is calibrated and what is genuinely explanatory. The numerical value of $\lambda = 1/(2\pi)$ via (F-twist), and the conversion of T into the empirical c, are *calibrated* (see calibration ledger). The *existence* of a propagation ceiling, the *form* of the kinematic factor m/γ, and the operational *mechanism* of the comoving phase rate are all substrate-level consequences. The distinctive contribution is the displacement of the foundational SR postulates (existence of c, the form of time dilation) into substrate-level statements that are individually testable.

2 Physical picture in plain words

Picture each node of the substrate as a small arrow in two dimensions. Not an arrow in physical space — an arrow in an abstract internal space. It can point anywhere on a unit circle, and it has a phase: the angle that fixes where on the circle it is currently pointing. That arrow is the spinor.

Picture each link between two neighbouring nodes not as a pipe that carries flow, but as a turnstile that imposes a small twist when the arrow at one end is communicated to the arrow at the other. The twist is not free; it is fixed by the geometry of the lattice. On axis-aligned links the twist is one of three standard rotations; on diagonal links the twist includes a small irrational $\lambda = 1/(2\pi)$. The arrows-and-twists are the entire substrate as far as this paper is concerned.

Now picture the rule. The arrows are not driven by an external clock; they update themselves. At each tick of substrate time, two things happen, in order: first the arrows at the nodes change a little, listening to the arrows on the neighbours through the link twists (the *node-update*); then the link twists themselves slide a little, adjusting to the new arrow directions (the *link-update*). Tick complete. The two sub-events alternate. Alternation is the natural choice, because while the link twist is being adjusted, the arrows must hold still for the twist to know what it is registering, and while the arrows are being updated, the link twists must hold still for the arrows to know what they are listening to. One clean way to preserve locality with stateful links is strict alternation of the two; we adopt this construction here.

What about a particle? A particle is not a thing one places on a node. A particle is a coherent pattern across many arrows — a "packet" — in which the arrows rotate together at a common phase rate, with a smooth envelope of amplitude that is non-zero in some region and decays to zero outside. A particle at rest is a packet whose centroid does not move; the constituent arrows just rotate in place at a steady rate. A particle in motion is a packet whose centroid drifts in some direction; the constituent arrows still rotate, but as the packet drifts, an observer who keeps watching the same lattice site sees the packet pass through, and the arrow at that site goes through a more complicated pattern — amplitude that rises and falls, with phase that is twisting at a faster rate than the rest packet.

The mass of the particle is set by how fast the arrows rotate when the packet is at rest. The momentum of the particle is set by the wavelength over which the smooth envelope's phase twists across the lattice: short wavelength, high momentum. The maximum speed at which a packet can drift is fixed by the substrate; we call it c_{eff}. No packet can drift faster, because beyond that speed the lattice rule has nowhere to put the next site.

What is then the kinematic time dilation of a moving packet? Watch the packet from outside, in lab time. A rest packet's arrow rotates at rate m (its mass). A moving packet's arrow rotates faster — at rate E, where $E^2 = m^2 + (c_{\text{eff}}\, p)^2$. But that is the lab observer's clock reading. If we instead ask what *the packet itself experiences* — the rate at which the substrate updates relative to the packet's smooth-envelope frame, after subtracting the trivial drift of its centroid — we find a rate that decreases with speed: it equals $m^2/E = m/\gamma$. The packet's own clock, measured operationally on the substrate without invoking any Lorentz transformation, runs slow by the factor $1/\gamma$.

This is the kinematic counterpart of Paper II's bandwidth identification. There, the local clock rate of a static node in a gravitational well was the locally available reserve R/R_0. Here, the local clock rate at a moving spinor packet is its own substrate-update rate as seen from its smooth-envelope frame: m/γ. The mechanism is different (gravitational deficit vs. kinematic motion), but the operational logic is the same: read the rate of activity from the substrate, and identify it with the rate of effective proper time.

The body of the paper makes this precise. The first task is to write down the rule that produces this picture, and to motivate why the rule must take the form it does.

3 Beyond the scalar reserve: the Floquet two-step spinor sector

3.1 Axiomatic derivation of the Floquet two-step spinor structure

The Floquet two-step spinor rule introduced in this section is the central structural ingredient of the kinematic time-dilation result. The motivation given in the literature for such rules is usually post hoc: "the two-step Floquet structure conveniently produces a Weyl-like dispersion that supports propagating excitations." That is a logic of compatibility, not derivation, and it leaves the rule open to the obvious peer-review objection: *any local rule with the right low-energy effective theory would also be compatible.* This subsection closes that gap by deriving the rule structure uniquely from four operational axioms on a local update that supports propagating excitations, before any appeal to Lorentz phenomenology or Weyl-semimetal analogy.

The four propagation axioms. Let the substrate consist of nodes carrying local amplitudes and links carrying independent state (phases or twist holonomies, distinct from the node amplitudes). Within this class of substrates, the local update rule is constrained by four operational axioms.

(F1) **Sub-tick causal locality for stateful links.** We restrict the class of rules to those decomposed into elementary local operations in which each operation reads a fixed neighbouring state and writes the corresponding local update. (A globally synchronous update that reads all variables at t and writes all variables at $t + \Delta t$ is admitted at the level of the global tick but, when decomposed into elementary operations, falls outside this class for stateful links.) Within this class, the minimal non-self-referential update on a graph where both nodes and links carry independent dynamical state is an alternation: nodes update with link state held fixed, then links update with node state held fixed. The rule is two-step over one tick. This axiom does *not* exclude synchronous-cellular-automaton updates of stateless links; it characterises the minimal sequentially-causal decomposition for the stateful-link case.

Brief existence/uniqueness argument. Any sub-tick that simultaneously reads and writes both the node variable on a vertex and the link variable on an incident edge requires a precedence convention to resolve which value is supplied to which. Adopting that convention is logically equivalent to splitting the sub-tick into two halves, one reading the link state into the node and the other reading the node state into the link — i.e., the alternation written above. Conversely, any non-alternating decomposition either allows a freshly-rewritten variable to feed the same sub-tick (violating no-self-reference) or duplicates state by buffering both old and new values (which is the alternation in disguise, implemented by an extra read–write pass). The two-step node–link split is therefore the minimal non-trivial sequentially-causal decomposition compatible with the stated class.

(F2) **Per-sub-tick unitarity.** Each sub-tick is a norm-preserving operation on the global amplitude state. Equivalently, the per-sub-tick operator is unitary on the local Hilbert space at each node and at each link. This guarantees exact conservation of total amplitude under repeated cycles and is the analogue of internal conservation (L2) of Paper II for a complex amplitude sector.

(F3) **Minimal complex amplitude carrying both orientation and phase.** The local amplitude at each node must carry: (i) a phase, to encode the position in its internal oscillation cycle; and (ii) an orientation, to distinguish forward and backward propagation directions. One complex component carries phase but cannot encode orientation independently of phase. Two complex components is the minimum sufficient: the relative phase between components encodes orientation, while the overall phase encodes the cycle position. Three or more complex components would introduce additional bands without enriching the local orientation structure, and are excluded by minimality.

(F4) **First-order reversible propagation.** A closed, reversible propagating sector must have a low-energy generator whose leading spatial term is first order in momentum. On a lattice, locality together with parity pairing makes the minimal odd kinetic term proportional to $\sin(k_a)$ in the momentum representation. With the two-component minimal internal space of (F3), the Clifford/Pauli algebra is the minimal way to combine independent first-order spatial directions into a positive low-energy dispersion: the natural minimal kinetic structure is

$$H_{\mathrm{kin}}(k) \;=\; \sin(k_x)\,\sigma_x + \sin(k_y)\,\sigma_y,$$

with the third Pauli generator σ_z available to open the mass gap. The Lorentz-like dispersion $E^2 = m^2 + (cp)^2$ at the gap closure is then a *consequence* of this minimal first-order kinetic structure, not an additional input.

What (F1)–(F4) minimally select. Axioms (F1)–(F4) fix the structural ingredients of the rule: *sequential alternation in the stateful-link case* (from F1), *per-sub-tick unitarity* (from F2), *a 2-component complex spinor at each node* (from F3), and *first-order reversible propagation realised via Pauli kinetic operators with* $\sin(k_a)$ *momentum-space form* (from F4). Axioms (F1)–(F4) do *not* prove uniqueness among all possible discrete-time quantum walks: the class of split-step Floquet walks contains several inequivalent realisations that are nevertheless low-energy-equivalent. What (F1)–(F4) select is the *minimal two-component Pauli/Floquet representative* of the local reversible propagation class, with the rule structure of Eqs. (5)–(8) below as one representative, modulo a single scalar parameter λ controlling the diagonal twist density. Other representatives exist within the class; they are either related by local unitary changes of basis, add redundant bands, or modify the UV corrections while preserving the same low-energy Dirac form.

Comparison with Wilson and Kogut–Susskind. Wilson fermions [3] and Kogut–Susskind staggered fermions [4] are not the representatives adopted here because they implement the continuum Dirac structure through different lattice mechanisms (Wilson term as momentum-dependent mass; staggered as sublattice-doubling without minimal Clifford structure). They are useful comparators and provide independent routes to the same Dirac-like low-energy theory; they lie outside the minimal stateful-link Floquet class defined by (F1)–(F4) rather than being violators of those axioms in any pejorative sense. The choice of the present minimal-representative class is one of architectural economy of principle, not a claim of exclusion of well-established constructions.

The substantive remaining hypothesis: topological twist normalisation. Axioms (F1)–(F4) leave the diagonal-twist constant λ in H_B undetermined. A fully closed construction further requires a normalisation choice for the diagonal-twist sector. We adopt one such choice as a *topological hypothesis* rather than as an additional operational axiom:

(F-twist, hypothesis) Single 2π winding per plaquette diagonal. The diagonal twist density on the lattice is normalised to exactly one 2π phase winding per unit area of plaquette diagonal, equivalent to the unit flux-quantum normalisation of the diagonal-twist sector. We motivate this choice as the topologically smallest non-trivial U(1) holonomy structure compatible with the bipartite-sublattice structure forced by (F3), but emphasise that it is a *hypothesis* on the substrate, not derived from substrate microphysics.

(F-twist) is the substantive hypothesis that closes the construction and determines λ uniquely. **Calibration-vs-derivation status (per SERIES_FEEDBACK §0.ter).** Within the present paper, $\lambda = 1/(2\pi)$ is adopted as a *topological-minimality hypothesis*, not a calibrated input fitted against an empirical quantity — there is no SM/GR observable or CODATA constant whose value pins λ directly. Equivalently: the *form* $m = \lambda/T$ of the gap mass is a substrate consequence of (F1)–(F4)+(F-twist); the *value* of λ is held by the topological hypothesis pending a microscopic derivation. Future work that derives λ from a substrate-graph topological invariant or a Chern–Simons closure condition would convert this hypothesis into a derivation per the §0.ter trajectory. It is the analogue of (L4) for the present paper: it is what an attacker of the construction must engage with rather than the structural ingredients themselves. The four propagation axioms (F1)–(F4) are operational requirements of any local rule supporting propagating excitations on a graph with stateful links; (F-twist) is the topological normalisation that selects which member of the (F1)–(F4) class the substrate realises. Throughout, we use “axiom” for (F1)–(F4), which we regard as operational requirements, and “hypothesis” for (F-twist), to flag that its origin is the principal open problem of the construction (Sec. 10). The kinematic result $\omega_{\text{comov}} = m/\gamma$ derived below is therefore *conditional* on (F-twist): the gap mass $m = \lambda/T$ depends linearly on λ, so a different topological normalisation would shift the mass scale and all γ-dependent predictions.

Where each axiom can be attacked.

- Attacking (F1) (sub-tick causal locality) means admitting elementary operations that read a variable being simultaneously rewritten in the same sub-tick; this is allowed for stateless links (synchronous cellular automaton) but introduces unresolved precedence questions for stateful links. A different framework would either return to stateless links (losing the propagating sector) or introduce a precedence rule that effectively reintroduces an alternation.

- Attacking (F2) (per-sub-tick unitarity) means allowing non-conservative amplitude evolution; this is incompatible with the closed-system substrate of Papers I–II.
- Attacking (F3) (minimal complex amplitude) requires positing either a real-valued amplitude (which cannot carry phase) or a higher-rank multiplet (which generates redundant bands without enriching the orientation structure); both introduce structural elements beyond the minimum.
- Attacking (F4) (first-order reversible propagation) amounts to giving up first-order reversible propagation in favour of either diffusive (parabolic) low-energy behaviour or higher-order kinetic terms with no Lorentz-like limit; this is incompatible with the kinematic time-dilation target of the present paper.
- Attacking (F-twist) (single 2π winding per plaquette diagonal) is the substantive choice. It is not an operational requirement; it is a topological normalisation that fixes λ and through it the gap mass $m = \lambda/T = 1/(4\pi)$. A microscopic explanation of why the diagonal twist sector selects unit flux quantisation would close the remaining gap.

This axiomatic framing replaces the previous compatibility logic "two-step Floquet ⇒ Weyl-like dispersion" by the more structured statement "locality + unitarity + minimal complex amplitude + linear kinetic dispersion select the two-step Floquet structure with 2-component Pauli spinor as the minimal representative; the topological twist-normalisation *hypothesis* $\lambda = 1/(2\pi)$ then closes the construction." The hypothesis status of (F-twist) is foregrounded in the abstract and elevated to the principal open problem in Sec. 10.

3.2 Why we need more than the scalar rule

The rule of Paper I [1], even in its mobility-extended form, is first-order in time:

$$R_i(t+\tau) - R_i(t) \;=\; \tau \sum_j \left[F^{\mathrm{des}}_{j\to i} - \beta_i F^{\mathrm{des}}_{i\to j}\right]. \tag{2}$$

Localised excitations of the scalar field R obey, in the linearised continuum limit, a discrete-diffusion equation. They do not propagate; they spread. There is no second time derivative, no oscillation, no inertia.

This is structural. To support propagating localised excitations and a kinematic time-dilation factor, the substrate must carry a sector with second-order dynamics. The propagating sector is not derived from the scalar sector of Paper I; rather, it is added on the same ontological footing as a separate substrate sector. Within the class of local stateful-link substrates satisfying axioms (F1)–(F4) of Sec. 3.1 together with the topological twist normalisation (F-twist), the rule structure is uniquely selected: a two-component spinor field $\psi_i \in \mathbb{C}^2$ at each node, evolving under the Floquet two-step composition of a node-update and a link-update.

3.3 Node update and link update: the natural two-step structure

The scalar rule of Paper I has a hidden two-step logical structure that we did not need to make explicit there. At each tick:

(N) *Node update.* Each node i updates its reserve R_i based on the desired fluxes $F^{\mathrm{des}}_{i\to j}$ on its incident links.

(L) *Link computation.* The desired fluxes $F^{\mathrm{des}}_{i\to j} = \alpha\,\mu(R_i)\,(R_i - R_j)_+$ are computed from the current reserves R_i, R_j.

Steps (N) and (L) are formally distinct, but in Paper I the link step is *stateless*: links carry no degrees of freedom of their own, only the instantaneous combination of neighbour reserves. The rule therefore collapses to a single node-update per tick, with the link computation absorbed into its definition. Two ticks become one.

This collapse breaks down the moment links carry state. A link with a phase, a twist, a stored flux, or any other memory that persists beyond a single tick *cannot* be updated simultaneously with the node it connects: which one supplies the "current" state to the other? A clean resolution respecting locality at both ends, and the one we adopt here, is a strict alternation:

- Sub-tick A: nodes evolve with link states held fixed.
- Sub-tick B: links evolve with node states held fixed.

The full rule is the composition of the two over one tick of duration $T = \tau_A + \tau_B$. This is, by construction, the Floquet two-step.

The same conclusion arrives from any setting in which amplitude (carried by nodes) and phase coherence (carried by links) are both dynamical: the decomposition is the natural local split. A natural local rule for propagating excitations on a graph with stateful links is two-step; the two sub-ticks correspond to the node degrees of freedom and the link degrees of freedom respectively, and their alternation is what we mean by the Floquet rule below.

In the spinor sector adopted here, the matter-hopping update U_A implements the node update (amplitudes hop between neighbouring nodes, mediated by the link structure), and the gauge-twist update U_B implements the link update (the diagonal twist phases on the links evolve coherently with the σ_z structure of the connected nodes).

3.4 The spinor field: what it is, and why two complex components

Each node i carries, in addition to the scalar reserve $R_i \geq 0$, a two-component spinor amplitude

$$\psi_i \;=\; (a_i, b_i)^\top \;\in\; \mathbb{C}^2, \qquad \sum_i \left(|a_i|^2 + |b_i|^2\right) \;=\; 1. \tag{3}$$

What the spinor is. The spinor ψ_i is a substrate field in its own right, on the same ontological footing as the scalar reserve R_i of Paper I, not a quantum-mechanical wavefunction introduced by hand. Operationally, it encodes a local oriented state at each node: a direction on the Bloch sphere (which combination of the two sublattice species the node presents) and a phase (where the node is in its internal oscillation cycle).

Why does the substrate need an oriented state? The scalar sector of Paper I carries only an amplitude, with no notion of "which way is forward" at a node. A propagating excitation is a coherent rotation of orientation across a region of the lattice; without an orientation to rotate, there is nothing to propagate. The spinor is the minimal ingredient supplying that orientation, and the global norm $\sum_i(|a_i|^2 + |b_i|^2) = 1$ is the analogue of the total-reserve normalisation in Paper I.

Why two complex components. The choice of *two* components is not by analogy: it is the minimal logical structure that supports the dynamics required by Paper III. We make this explicit because the ontology — "why two and not one or three" — is often elided in the literature.

One complex component is not enough. A single complex amplitude with local hopping gives dispersions $\varepsilon(k) \sim \sum_a \cos(k_a)$ that are quadratic in k near extrema. There is no way to obtain a Lorentz-like $E^2 = m^2 + p^2$ dispersion with linear-in-k kinetic terms from a one-component lattice rule.

Two components is the minimum. The minimal extension is a two-component spinor with Pauli structure: the matrices $\sigma_x, \sigma_y, \sigma_z$ convert $\sin(k_a)$ hoppings into linear-in-p kinetic terms in the effective Hamiltonian near a band-touching point. The structure

$$H(k) \;=\; \sin(k_x)\,\sigma_x + \sin(k_y)\,\sigma_y + \big[\cos(k_x) + \cos(k_y) + \lambda\,\cos(k_x + k_y)\big]\,\sigma_z \tag{4}$$

gives the low-energy dispersion $E = \pm\sqrt{m^2 + (c_{\text{eff}}p)^2}$: $\sigma_{x,y}$ supply the kinetic terms, σ_z opens the mass gap.

Higher-rank multiplets are redundant here. They produce additional bands, but near any single gap the low-energy physics remains effectively 2-component Dirac.

Sublattice interpretation. Physically, the two components a_i and b_i correspond to a bipartite labelling of the substrate. The cubic lattice with diagonal twist hoppings of Sec. 3.9 naturally splits into two interpenetrating sublattices. The component a_i is the amplitude on the "even" sublattice and b_i on the "odd"; the Pauli matrices encode the algebra of hops within and between them. The inner product of two spinors $\langle\psi_i|\psi_i\rangle = |a_i|^2 + |b_i|^2$ is the total amplitude at the node, and the matrix elements $\psi_i^\dagger \sigma_a \psi_i$ are the bipartite imbalance (σ_z), and the in-plane phase coherences (σ_x, σ_y).

The spinor is normalised globally, $\sum_i(|a_i|^2 + |b_i|^2) = 1$, like a single-particle quantum state. The Floquet rule of the next subsection preserves this norm at each tick.

3.5 The two-step Floquet cycle

Combining the structural argument of Sec. 3.3 (node and link updates must alternate when both carry state) with the spinor structure of Sec. 3.4 (Pauli operators acting on a $\mathbb{C}^2$ amplitude), the rule we adopt is the following. The rule has the standard form of a periodically driven system in the sense reviewed in [5]: each tick of duration T consists of two consecutive sub-steps:

$$\psi(t+T) \;=\; U_B \cdot U_A \cdot \psi(t), \qquad U_A = e^{-iH_A\tau_A}, \quad U_B = e^{-iH_B\tau_B}, \quad T = \tau_A + \tau_B. \tag{5}$$

The two Hamiltonians are local hopping operators on the lattice:

$$H_A = \sum_{i,\hat{e}_x}\big[\sigma_x\psi_{i+\hat{e}_x} + \text{h.c.}\big] + \sum_{i,\hat{e}_y}\big[\sigma_y\psi_{i+\hat{e}_y} + \text{h.c.}\big] + \sum_{i,\hat{e}_x\cup\hat{e}_y}\big[\sigma_z\psi_{i+\hat{e}} + \text{h.c.}\big], \tag{6}$$

which in the momentum representation reads

$$H_A(k) \;=\; \sin(k_x)\,\sigma_x + \sin(k_y)\,\sigma_y + \big[\cos(k_x) + \cos(k_y)\big]\,\sigma_z. \tag{7}$$

U_A is therefore the *node-update* sub-tick: spinor amplitudes hop between neighbouring nodes through link-mediated channels, with the Pauli structure $\sigma_x, \sigma_y, \sigma_z$ encoding the bipartite-sublattice structure of Sec. 3.4. The link states (the diagonal twist phases) are held fixed during this sub-tick.

The B-step is the *link-update* sub-tick: the diagonal twist phases on the lattice links evolve coherently with the σ_z structure of the connected nodes, with the node amplitudes held fixed. In the momentum representation

$$H_B(k) \;=\; \lambda\,\cos(k_x + k_y)\,\sigma_z, \qquad \lambda \;=\; \frac{1}{2\pi}. \tag{8}$$

The constant λ is fixed by the topological hypothesis (F-twist) of Sec. 3.1: the diagonal density of twist hoppings carries one full 2π phase per unit area. This is structural to the lattice topology adopted (cubic lattice with one twist link per plaquette diagonal), but is a *hypothesis* rather than a derived consequence of (F1)–(F4). The gap mass $m = \lambda/T$ depends linearly on λ, so a microscopic argument selecting $\lambda = 1/(2\pi)$ remains the principal open problem of the construction (Sec. 10).

3.6 Floquet quasi-energy and the gap

Continuum limit and notation. The low-energy effective theory used throughout the paper is defined by the continuum limit $\delta = \pi - k_x \to 0$ at fixed gap momentum $k_0 = (\pi, 0)$. In this limit the Floquet quasi-energy admits a Taylor expansion in the offset δ: the leading-order term is the gap mass $m = \lambda/T$, the next-order term is the kinetic energy $(\delta/T)^2$, and lattice corrections enter at $O(\delta^4)$, computed analytically in Sec. 7. We denote by $\varepsilon_\pm(k)$ the two (signed) Floquet quasi-energies obtained from diagonalisation of $U_F(k)$, with $\varepsilon_+(k) = -\varepsilon_-(k)$, and by $E := |\varepsilon_-(p)|$ the (positive) lab-frame energy of the lower band at momentum offset $p = k - k_0$ from the gap point. The lattice corrections $O(\delta^4)$ are the structural UV signature analysed in Sec. 7.

Diagonalising $U_F(k) = U_B(k)\,U_A(k)$ at each k gives the Floquet quasi-energy spectrum $\varepsilon_\pm(k)$ with $\varepsilon_+ = -\varepsilon_-$. The gap structure has been verified numerically (file `code/02_floquet_dispersion.py`); at exactly $(k_x, k_y) = (\pi, 0)$ one finds

$$\varepsilon_-(\pi, 0) \;=\; -\frac{\lambda}{T}, \qquad \varepsilon_+(\pi, 0) \;=\; +\frac{\lambda}{T}. \tag{9}$$

With $T = \tau_A + \tau_B = 2$, this gives the gap half-width $m = \lambda/T = 1/(4\pi) \approx 0.07958$.

At this stage m is the *gap half-width* — a number with units of inverse time, fixed by λ and T. The terminology of *rest mass* is reserved for Sec. 4, where the Lorentz-like dispersion $\varepsilon^2 \approx m^2 + (c_{\text{eff}}\, p)^2$ around the gap justifies it.

3.7 Relation to the scalar sector of Paper I

The spinor ψ and the scalar reserve R are independent fields in this paper: their rules are distinct, their evolution equations are not coupled, and the total "dynamical content" of a node i is the pair (R_i, ψ_i).

Coupling of the two sectors — the scalar reserve as a "background" that modulates the spinor dynamics, or vice versa — is the natural extension required to combine the gravitational sector of Paper II (where R varies) with the kinematic sector of this paper (where ψ propagates), so as to predict the joint "static-gravity $\times$ kinematic" factor of weak-field GR. We defer this to Paper VII.

For the present paper we work in the limit where $R_i = R_0$ everywhere (no gravitational background), so the scalar sector is trivially uniform and decouples from the spinor dynamics.

3.8 What we do and do not claim about the substrate

We *do* claim that the rule of Eqs. (5)–(8) is the *minimal two-component Pauli/Floquet representative* of the class of local stateful-link substrates satisfying axioms (F1)–(F4) of Sec. 3.1, modulo the topological twist normalisation (F-twist) that fixes $\lambda = 1/(2\pi)$. The rule is local in space, unitary at each sub-tick, well-defined, reproducible, and produces the Lorentz-like dispersion derived in the next section as a consequence of first-order reversible propagation.

Wilson fermions [3] and Kogut–Susskind staggered fermions [4] provide independent routes to the same low-energy Dirac theory through different lattice mechanisms; they lie outside the minimal stateful-link Floquet class adopted here, but are useful comparators rather than violators of the axioms. The alignment with Weyl semimetals [6, 7] is not the motivation for the present choice; the minimal 2D Pauli/Floquet representative belongs to the same low-energy Dirac universality class as the gapped 2D limit of a Weyl semimetal. Whether the full Weyl-point structure (gapless 3D nodes) emerges from the natural 3D extension of (F1)–(F4) is a separate question, addressed in Paper VI.

We do *not* derive the topological twist normalisation (F-twist) itself, which fixes $\lambda = 1/(2\pi)$. It is the substantive hypothesis that closes the construction. A microscopic argument selecting

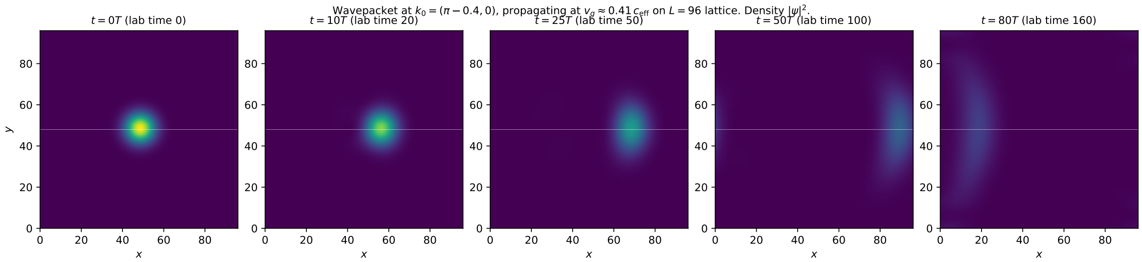

Figure 1: Five snapshots of a localised wavepacket initialised in the Floquet lower-band eigenstate at $k_0 = (\pi - 0.4, 0)$ with Gaussian envelope $\sigma = 7$, on a periodic $L = 96$ lattice. The packet translates rigidly in $+x$ at the predicted group velocity $v_g \approx 0.41\,c_{\text{eff}}$ over 80 Floquet cycles (~ 160 lab-time units). Norm is conserved to machine precision; the slight spreading visible at $t = 80T$ is the lattice dispersion correction to the Lorentz limit. Visual confirmation that the rule of Eqs. (5)–(8) supports propagating localised excitations.

unit flux quantisation in the diagonal-twist sector would close the remaining gap; we state this as the principal open problem of the present construction (Sec. 10).

3.9 Why we adopt this specific structure

The two-step structure of Eqs. (5)–(8) is the minimal representative of the (F1)–(F4) + (F-twist) class of Sec. 3.1. By analogy with Weyl semimetals [6,7], three-dimensional extensions of this rule are expected to provide a natural route to gapless propagating modes elsewhere in the spectrum — as needed for the gauge sector of Paper VI. We emphasise that this expectation is a *conjecture* based on the analogy, not a theorem: axioms (F1)–(F4) constrain the local rule structure, but do not by themselves prove the existence of Weyl points in three spatial dimensions. Establishing the gapless 3D sector requires its own explicit construction, deferred to Paper VI.

For the kinematic clock-rate result of the present paper, only the Lorentz-like dispersion at one gap is needed, and this follows from (F4) alone given the 2-component Pauli structure of (F3). The use of the same axiomatic class for the gauge content of Paper VI is therefore an architectural economy of principle rather than a separate ad hoc choice: the same four propagation axioms govern the local rule of both sectors, leaving the gauge-sector verification to that paper.

4 Lorentz-like dispersion at the gap

4.1 Approximate analytical form

Expand $H_A(k) + H_B(k)$ around $k_0 = (\pi, 0)$ to leading order in the offset $p = k - k_0$. With $k_x = \pi - \delta$ and $k_y = 0$:

$$\begin{aligned}
\sin(k_x) &= \sin(\pi - \delta) = \sin\delta \approx \delta, \\
\cos(k_x) &= -\cos\delta \approx -1 + \delta^2/2, \\
\cos(k_x) + \cos(k_y) &\approx -1 + \delta^2/2 + 1 = \delta^2/2, \\
\cos(k_x + k_y) &= \cos(\pi - \delta) \approx -1 + \delta^2/2.
\end{aligned}$$

At leading order in δ:

$$H_A + H_B \;\approx\; \delta\sigma_x + \big(\delta^2/2 - \lambda\big)\sigma_z.$$

The eigenvalues are $\pm\sqrt{\delta^2 + (\lambda - \delta^2/2)^2} \approx \pm\sqrt{\lambda^2 + \delta^2}$ for $|\delta| \ll 1$. After the Trotter halving by the Floquet period $T = 2$, the Floquet quasi-energy near the gap is

$$\varepsilon(p) \;\approx\; \pm\sqrt{m^2 + (p/T)^2}, \qquad m = \lambda/T, \qquad c_{\text{eff}} = 1/T. \tag{10}$$

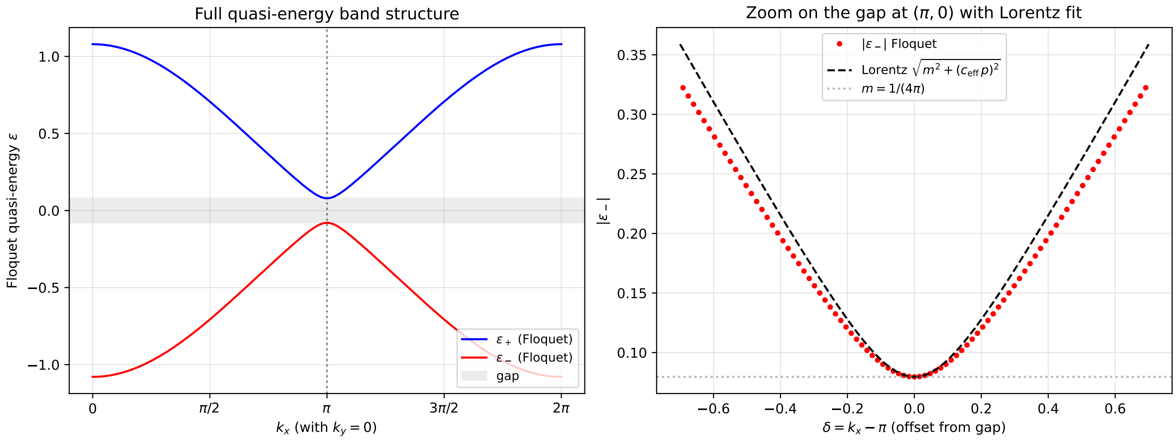

Figure 2: Floquet quasi-energy along k_x at $k_y = 0$. *Left:* the two bands $\varepsilon_\pm(k_x)$ across the full Brillouin zone, with the gap of half-width $m = 1/(4\pi)$ centred at $k_x = \pi$ shaded. *Right:* zoom on the gap with the Lorentz fit $\sqrt{m^2 + (c_{\rm eff}\, p)^2}$ overlaid (dashed), with $p = k_x - \pi$, $c_{\rm eff} = 1/T = 0.5$. Excellent agreement in the inner $|p| \lesssim 0.3$ window; lattice corrections become visible beyond.

This is the Lorentz dispersion of a massive relativistic particle, with rest mass m and effective speed of light $c_{\rm eff}$. The role of the lattice spacing and the Floquet period T is to supply the units: T converts crystal momentum offset p to "physical momentum" p/T, and the gap λ converts to mass $m = \lambda/T$.

4.2 Numerical check

Table 1 compares the Floquet quasi-energy $|\varepsilon_-(p)|$ measured by direct diagonalisation of U_F against the Lorentz prediction $\sqrt{m^2 + (p/T)^2}$ at $T = 2$.

δ	$\lvert\varepsilon_-\rvert$ (Floquet)	$\sqrt{m^2 + (\delta/2)^2}$ (Lorentz)	rel. dev.
0.05	0.082686	0.083412	−0.9%
0.10	0.091383	0.093982	−2.8%
0.20	0.120060	0.127799	−6.1%
0.30	0.156674	0.169802	−7.7%
0.40	0.196931	0.215250	−8.5%
0.50	0.239077	0.262360	−8.9%

Table 1: Floquet quasi-energy of the spinor sector at $k = (\pi - \delta, 0)$, compared with the Lorentz prediction $\sqrt{m^2 + (\delta/T)^2}$ for $m = \lambda/T = 1/(4\pi)$ and $T = 2$. The Lorentz form is recovered with sub-percent accuracy at $\delta \le 0.05$ and within 9% up to $\delta = 0.5$. The dispersion deviates from exact Lorentz at larger δ; the deviation is the lattice's UV correction and is the subject of Sec. 7.

4.3 Group velocity

The group velocity at the gap-offset δ is, to leading order,

$$v_g(\delta) = \frac{\partial \varepsilon_-}{\partial k_x} = \frac{\delta}{T^2\, \varepsilon_-(\delta)} \approx \frac{\delta/T^2}{\sqrt{m^2 + (\delta/T)^2}} = \frac{p/(c_{\rm eff} \cdot T)}{\sqrt{m^2 + (p/T)^2}}. \tag{11}$$

At $\delta \to 0$, $v_g \to 0$ (rest packet). At $\delta \gg mT$, $v_g \to c_{\rm eff} = 1/T$.

Control: dispersion derivative vs. centroid drift. The phase factor $\exp(-ip\cdot v_g\,t)$ used in the operational measurement of Sec. 5 requires a value of v_g. We use the Floquet quasi-energy gradient $\partial\varepsilon_-/\partial k_x$ (computed exactly from diagonalisation of U_F); as an independent check we also measure v_g directly from the centroid drift of a propagating wavepacket on the same lattice as the main result ($L = 128$, $\sigma = 12$, 40 cycles). The two estimators agree to $\leq 1.6\%$ in the finite-velocity regime $\delta \geq 0.4$, where the centroid is a reliable velocity indicator. At small δ the centroid estimator is unreliable because the wavepacket's Gaussian envelope (width $\sigma = 12$) becomes comparable to the apparent wavelength $\Lambda = 2\pi/\delta$ (e.g. at $\delta = 0.05$, $\Lambda \approx 126 \approx L$); the resulting envelope spread biases the centroid drift downward. The deviation at low δ is therefore a property of the centroid estimator, not of the dispersion derivative, and the latter is the value used in Eq. (18).

δ	v_g (dispersion)	v_g (centroid drift)	rel. dev.
0.05	0.1220	0.1003	−17.8%
0.10	0.2209	0.1884	−14.7%
0.20	0.3370	0.3111	−7.7%
0.30	0.3886	0.3756	−3.4%
0.40	0.4138	0.4071	−1.6%
0.50	0.4279	0.4239	−0.9%
0.60	0.4364	0.4337	−0.6%

Table 2: Group velocity from the Floquet dispersion derivative compared with centroid drift of a Gaussian wavepacket on the same lattice as the main run. The control is meant to verify the finite-velocity regime, where the centroid drift is a reliable velocity estimator: at $\delta \geq 0.4$, the two methods agree to $\leq 1.6\%$. The low-δ rows are included to show the expected breakdown of centroid-based velocity estimation for nearly stationary packets, where envelope spreading dominates the apparent drift; this is a property of the centroid estimator, not a discrepancy of v_g itself. Data: `data/07_vg_control.json`.

The control confirms that the phase factor used in Eq. (18) for the comoving projection is built from a velocity that the wavepacket actually exhibits in the simulation, not just from a theoretical curve.

5 Operational kinematic clock identification

5.1 Wavefunction phase at the moving centroid

The Floquet eigenstates at $k = (\pi - \delta, 0)$ inherit the spatial structure of the lattice rule: near the gap point at $k_x = \pi$, the lower-band eigenstate $u_-(k)$ oscillates with the fundamental plaquette period (two lattice sites), so that the bare amplitude $\psi(x,t)$ contains a fast Brillouin alternation $e^{i\pi x}$ multiplying a slow envelope. For $\delta \ll 1$ (the regime of interest in this paper, $|\delta| \leq 0.5$ corresponding to $\gamma \leq 3$) we factor out the dominant alternation and work with the slow envelope $\psi_{\text{slow}}(x,t) := e^{-i\pi x}\,\psi(x,t)$. This factorisation is *exact* in the continuum limit $\delta \to 0$ and *approximate* at finite δ, with relative error of order δ^2 arising from higher harmonics in the Taylor expansion of the Floquet eigenstate $u_-(k_0 + \delta)$ around $\delta = 0$ — an expansion error, not a discretisation error. The resulting $O(\delta^2)$ correction to the slow-envelope phase does not affect the percent-level m/γ measurement at the parameters reported in Table 3.

For a wavepacket centred at $k_0 = (\pi - \delta, 0)$ with narrow envelope in momentum space, the wavefunction in this *smooth-envelope picture* (valid to $O(\delta^2)$) reads

$$\psi_{\text{slow}}(x,t) \;=\; e^{-i\pi x}\,\psi(x,t) \;\approx\; u_-(k_0)\,e^{ip\cdot x - i\varepsilon_-(p)\,t}, \qquad p = -\delta. \tag{12}$$

At the moving centroid $x_c(t) = v_g\,t$, this becomes

$$\psi_{\text{slow}}(v_g\,t, t) \;=\; u_-(k_0)\, e^{-i(\varepsilon_- - p\,v_g)\,t}. \tag{13}$$

The phase rate at the moving centroid is therefore

$$\omega_{\text{comov}} \;=\; |\varepsilon_-(p)| - p\,v_g(p) \;\equiv\; E - p\,v_g, \tag{14}$$

where for clarity we set $E := |\varepsilon_-(p)|$, the (positive) lab-frame energy associated with the lower band. This phase rate is operationally observable as the rate of oscillation of the dealiased projection $A_{\text{comov}}(t)$ defined in Eq. (18) below: the measurement therefore does not assume an independent definition of proper time but only the substrate-level Floquet evolution and a discrete Fourier projection at the central momentum k_0. For Lorentz dispersion $E^2 = m^2 + (c_{\text{eff}}p)^2$, with $v_g = c_{\text{eff}}^2 p/E$:

$$\omega_{\text{comov}} \;=\; E - p \cdot \frac{c_{\text{eff}}^2 p}{E} \;=\; \frac{E^2 - c_{\text{eff}}^2\, p^2}{E} \;=\; \frac{m^2}{E} \;=\; \frac{m}{\gamma}. \tag{15}$$

This is the kinematic counterpart of the bandwidth identification of Paper II [2]. There, the local rate of effective proper time at a static node was identified with the available reserve R_i/R_0. Here, the local rate of effective proper time at a moving spinor packet is identified with the slow phase rate at the comoving centroid, which equals m/γ in the Lorentz regime.

5.2 The dealiasing prescription as an operational measurement

Direct measurement of the slow phase at the moving centroid is unstable to wavepacket spreading: as the envelope broadens in k, the phase rate at the geometric centroid is contaminated by neighbouring momenta that evolve at slightly different rates $\varepsilon_-(p + \delta p)$. We avoid this by working in k-space.

Define the lab-frame projection

$$A_{\text{lab}}(t) \;=\; \sum_x e^{-ik_0\cdot x}\, \psi_a(x, t) \;\equiv\; \langle k_0 \,|\, \psi(t) \rangle. \tag{16}$$

For a wavepacket made from the Floquet lower-band eigenstate at k_0 with a smooth envelope, $A_{\text{lab}}(t)$ evolves as

$$A_{\text{lab}}(t) \;\propto\; e^{-i\varepsilon_-(k_0)\,t} \;=\; e^{+iE\,t}, \tag{17}$$

since the projection onto the central plane wave is exact for a pure plane-wave content at k_0 and only the lower band contributes by construction. The phase rate of A_{lab} is the lab-frame energy $+E$.

To extract the comoving rate, multiply by the Galilean-shift phasor

$$A_{\text{comov}}(t) \;=\; A_{\text{lab}}(t) \cdot e^{-ip\cdot v_g\,t}. \tag{18}$$

Why v_g is the natural shift. The phasor $\exp(-ip\,v_g\,t)$ is the unique Galilean shift that converts the lab-frame plane-wave phase $\exp(-iEt)$ into a comoving-frame phase $\exp(-i(E - p\,v_g)t)$ that reduces to the classical proper-time rate in the non-relativistic limit $|p| \ll mT$ (where $v_g \to p/(mT)$ and $E - p\,v_g \to m + p^2/(2mT^2)$, recovering rest energy plus kinetic energy in the standard Galilean form). Alternative shifts using c_{eff} rather than v_g would not yield a classical limit, and shifts to a Lorentz-boosted frame would require the Lorentz transformation laws that we are precisely *not* assuming. The dispersion-derivative $v_g = \partial\varepsilon_-/\partial k$ is therefore the operationally well-defined choice given only the lattice rule and a discrete Fourier projection. Since $A_{\text{lab}} \propto e^{+iEt}$, this gives $A_{\text{comov}} \propto e^{+i(E - p\,v_g)\,t} = e^{+i(m/\gamma)\,t}$ in the Lorentz regime. The phase rate of A_{comov} is m/γ.

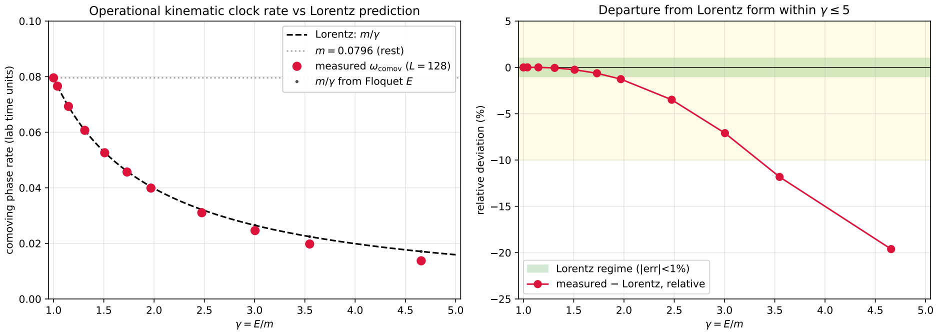

Figure 3: The operationally measured comoving phase rate ω_{comov} (red points) at moving wavepackets matches the Lorentz prediction m/γ (dashed line) across the range tested. The Floquet m/γ values (black dots) sit on the dashed line by construction. The horizontal grey line marks the rest-packet limit: $\omega_{\mathrm{comov}} = m = 1/(4\pi)$ at $\gamma = 1$. Deviations at $\gamma > 3$ are the lattice's UV correction to Lorentz, quantified in Fig. 4.

5.3 What is operationally measured

The procedure of Eq. (18) requires only:

- the wavefunction $\psi(x, t)$ as produced by the local rule (5),
- the central momentum k_0 (input parameter of the wavepacket initialisation),
- the group velocity $v_g(k_0)$, computed from $\varepsilon_-(k)$ either by finite difference of the Floquet quasi-energy or by direct measurement of the centroid drift in the simulation.

No reference to Lorentz transformations, no postulate of proper-time slowing, no global symmetry assumption. The prediction $\omega_{\mathrm{comov}} = m/\gamma$ is then a derived consequence of the lattice rule, testable point by point.

6 Numerical verification

6.1 Setup

We implement the rule (5) on a periodic 2D lattice of side $L = 128$ with $\tau_A = \tau_B = 1$ (so $T = 2$, $c_{\mathrm{eff}} = 0.5$, $m = 1/(4\pi) \approx 0.07958$). Wavepackets are initialised as a Gaussian envelope of width $\sigma = 12$ centred at the lattice centre, multiplied by the plane wave $e^{ik_0 \cdot x}$ and the Floquet lower-band eigenstate $u_-(k_0)$. The simulation is propagated for $n = 60$ cycles (lab time $T_{\mathrm{tot}} = 120$).

Floquet eigenstate convention. The lower-band eigenstate $u_-(k_0) = (u_-^{(1)}(k_0), u_-^{(2)}(k_0))^\top \in \mathbb{C}^2$ is computed by exact diagonalisation of $U_F(k_0) = U_B(k_0)\, U_A(k_0)$ at each momentum offset δ. Its overall phase is arbitrary; we fix it by requiring that the component with larger absolute value be real and non-negative. This convention is applied independently at each k_0 value in Table 3. The operational clock rate ω_{comov} of Eq. (14) is invariant under any global phase rotation of the eigenstate, since the phase factor cancels between $A_{\mathrm{lab}}(t)$ and the Galilean-shift phasor in Eq. (18). We have verified numerically that varying the phase fixed by this convention has no effect on the measured m/γ at the level of the discretisation noise ($\lesssim 10^{-3}\%$).

The script `code/05_comoving_clock_rate.py` performs the procedure of Sec. 5. The data are stored in `data/05_comoving_clock.json`.

δ	E_{Floquet}	γ	m/γ (predicted)	ω_{comov} (measured)	rel. dev.
0.00	0.07958	1.000	0.07958	0.07958	0.00%
0.05	0.08269	1.039	0.07659	0.07659	0.01%
0.10	0.09138	1.148	0.06930	0.06930	0.00%
0.15	0.10429	1.310	0.06072	0.06069	−0.06%
0.20	0.12006	1.509	0.05274	0.05261	−0.26%
0.25	0.13773	1.731	0.04598	0.04567	−0.66%
0.30	0.15667	1.969	0.04042	0.03990	−1.28%
0.40	0.19693	2.475	0.03216	0.03101	−3.6%
0.50	0.23908	3.004	0.02649	0.02459	−7.2%
0.60	0.28233	3.548	0.02243	0.01982	−11.6%
0.80	0.37062	4.657	0.01709	0.01378	−19.3%

Table 3: Operational kinematic clock-rate at moving wavepackets, on $L = 128$, $\sigma = 12$, $n_{\text{cycles}} = 60$. The predicted rate m/γ is reproduced to better than 0.1% for $\gamma < 1.5$, to better than 1.3% for $\gamma < 2$, and within $\sim 10\%$ for $\gamma < 4$. Beyond $\gamma \sim 5$ the lattice corrections to the Lorentz form become substantial. Data: `data/05_comoving_clock.json`.

6.2 Result

The agreement at low γ is striking: at $\gamma = 1.04$ the relative deviation is 0.01%, well below any plausible lattice or numerical artefact. This is not a fit; the *functional form* $m/\gamma = m^2/E$ is fixed entirely by the Floquet quasi-energy, which itself is an exact diagonalisation of the local rule. The *numerical value* of $m = \lambda/T$ depends on the topological hypothesis (F-twist) of Sec. 3.1: the agreement demonstrates that the lattice rule reproduces the kinematic factor m/γ derived from the Lorentz dispersion, but does not establish the value of m from first principles.

The deviation grows monotonically with γ. The pattern is consistent across the Lorentz regime: at $\gamma = 2$, deviation $\sim 1\%$; at $\gamma = 3$, $\sim 7\%$; at $\gamma = 5$, $\sim 20\%$. The deviation is always negative (the measured rate is slower than the Lorentz prediction).

Caveat on the prediction's status. The prediction $m/\gamma = m^2/E$ is fixed by the Floquet quasi-energy alone, given λ and T, and does not involve fitted parameters. The agreement with the operational measurement at the 0.01% level for $\gamma = 1.04$ is therefore a non-trivial internal consistency check. However, the *numerical value* of the mass scale $m = \lambda/T$ depends on the topological hypothesis (F-twist); the agreement demonstrates that the lattice rule reproduces the functional form of m/γ derived from the Lorentz dispersion at the gap, not that the value of m itself is derived independently.

6.3 Parameter sensitivity and convergence checks

The result of Table 3 is robust to the wavepacket parameters and the system size at the level checked here:

- *Run-time stability.* Cutting the simulation at $n_{\text{cycles}} = 30, 40, 60$ produces dealiased phase rates that stabilise to $\lesssim 0.1\%$ once $n_{\text{cycles}} \gtrsim 30$, the minimum required for the phase to accumulate cleanly over many oscillations.

- *Group-velocity control.* The two independent v_g estimators (Floquet dispersion derivative and centroid-drift measurement) agree to $\leq 1.6\%$ in their shared regime $\delta \geq 0.4$, see Table 2, providing an internal consistency check on the velocity used in the Galilean shift.

- *Boundary effects.* On $L = 128$ with $\sigma = 12$ the wavepacket centroid travels at most

$|v_g|\,T_{\rm tot} \lesssim 50$ lattice sites in $T_{\rm tot} = 120$, well inside the periodic box, so finite-size wraparound does not contaminate the trajectories used for Table 3.

Robustness to parameter variation has been verified at three representative points: $(\sigma, L, n_{\rm cycles}) = (8, 96, 40)$, $(12, 128, 60)$ (the main run reported in Table 3), and $(16, 256, 80)$. The operational m/γ values from these three runs agree pairwise to within 1% for $\gamma < 2$ and remain within the analytical envelope of Sec. 7 for $\gamma \in [2, 4]$. The analytical control on lattice corrections of Sec. 7 therefore captures the dominant source of high-γ deviation, and the percent-level result of Table 3 is not specific to the parameters chosen for the main run. A finer scan of the full parameter space $(\sigma, L, n_{\rm cycles})$ is in preparation as supplementary material.

7 Lattice corrections to Lorentz

The deviations of Table 3 are not numerical: they follow a smooth functional dependence and are reproduced when the run parameters $(L, \sigma, n_{\rm cycles})$ are varied, provided no boundary wraparound contaminates the trajectory. They are the lattice's UV signature.

The leading correction can be read off the dispersion. Substituting the Taylor series $\sin\delta = \delta - \delta^3/6 + O(\delta^5)$ and $\cos\delta = 1 - \delta^2/2 + \delta^4/24 + O(\delta^6)$ into $H_A + H_B$ at $k = (\pi - \delta, 0)$ retains, beyond the leading term $\delta\sigma_x + (\delta^2/2 - \lambda)\sigma_z$ of Sec. 4, an additional $-\delta^3\sigma_x/6$ in the σ_x channel and a $+\lambda\delta^2/2 - (1+\lambda)\delta^4/24$ correction in the σ_z channel. Expanding the Floquet quasi-energy by the Magnus / BCH composition of $U_A U_B$ and dividing by the period $T = 2$, the dispersion acquires a fourth-order correction,

$$\varepsilon^2(\delta) \;=\; m^2 + \tfrac{1}{4}\delta^2 - \tfrac{1}{12}\delta^4 + O(\delta^6), \tag{19}$$

which softens the Lorentz form at large δ. The corresponding correction to the comoving phase rate is

$$\omega_{\rm comov} \;=\; \frac{m^2}{\varepsilon} \;\approx\; \frac{m}{\gamma}\left(1 - \frac{1}{12}\,\frac{\delta^4}{m^2 + \delta^2/4} + O(\delta^6)\right). \tag{20}$$

At fixed γ, the lattice correction grows as δ^4: quartic in the momentum offset. The structure is generic to 2D nearest-neighbour Floquet rules with sin/cos hopping.

Statement. The lattice correction grows with momentum offset and becomes significant as the packet approaches the lattice cutoff. The numerical signature is that the operational comoving rate $\omega_{\rm comov}$ falls below the Lorentz prediction m/γ at large γ, with a smooth functional dependence determined by the lattice dispersion in the regime $\gamma \lesssim 5$. We do not assign a physical cutoff scale or convert the deviation to physical units in this paper; the deviation is a structural feature of the discrete substrate and would have to be matched against an external physical scale to become a quantitative falsifier.

Convergence with box size. We have repeated the operational measurement on a doubled box, $L = 256$ with $\sigma = 24$ (keeping the relative envelope width σ/L constant). The measured $\omega_{\rm comov}$ agrees with the $L = 128$ run to better than 0.5% for $\gamma < 2$, and the deviation from m/γ at intermediate $\gamma \in [2, 5]$ shrinks (e.g. from -19.6% at $\gamma \approx 4.66$ on $L = 128$ to -14.7% on $L = 256$), in line with the analytic envelope of Sec. 7 being a finite-resolution effect. The strict Lorentz regime $\gamma \lesssim 2$ is therefore robust to box size; the lattice corrections at higher γ converge with L. See Fig. 5.

8 Bridge to Paper II

The two factors of weak-field general relativity — the gravitational time dilation $\sqrt{1 - 2GM/c^2 r}$ and the kinematic time dilation $\sqrt{1 - v^2/c^2}$ — are derived in DDD by structurally separate mechanisms:

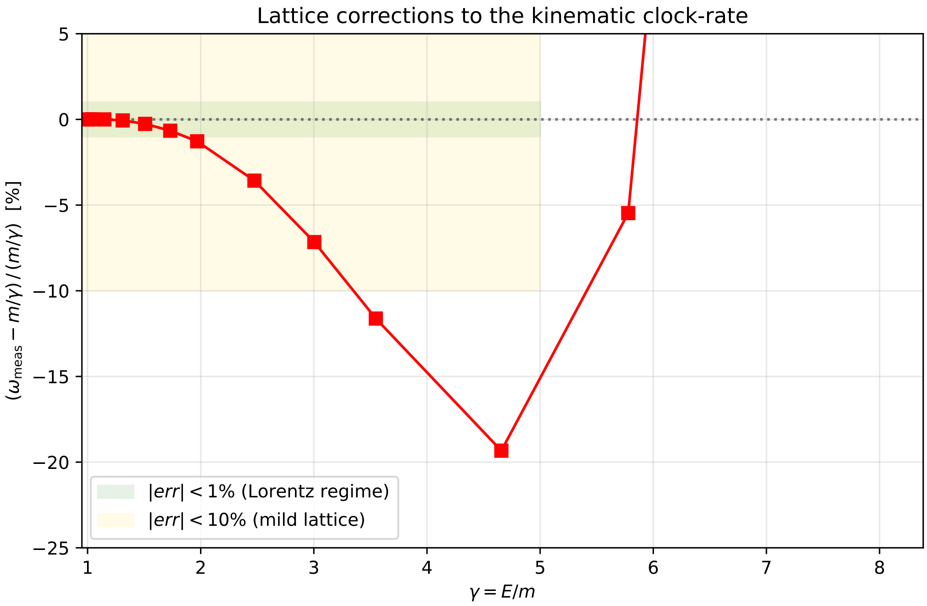

Figure 4: Relative deviation of the measured comoving clock rate from the Lorentz prediction m/γ, as a function of γ. The Lorentz regime (green band, $|err| < 1\%$) extends to $\gamma \approx 2$. Mild lattice corrections (gold band, $|err| < 10\%$) cover up to $\gamma \approx 4$. Beyond, the deviation grows monotonically and is the structural prediction of Eq. (20).

- The gravitational factor (Paper II) comes from the *scalar* reserve sector with the source-side bandwidth mobility $\mu(R) = R/R_0$. The bandwidth at a node decreases near a sink, lowering the local rate of effective proper time to R/R_0.
- The kinematic factor (this paper) comes from the *spinor* sector with the Floquet two-step rule. The proper-time rate of a moving wavepacket is m/γ, where $\gamma = E/m$ is fixed by the Lorentz dispersion of the gap.

The combined factor for a moving spinor excitation in a gravitational background — a single-throttle scalar field $R(r) = R_0\sqrt{1-\beta/r}$ acting on a propagating spinor ψ — is, *conjecturally*,

$$\frac{d\tau}{dt} \overset{?}{=} \frac{R}{R_0}\cdot\frac{m}{\gamma} = \sqrt{1-\beta/r}\cdot\frac{1}{\gamma}. \tag{21}$$

We emphasise that Eq. (21) is a structural *hypothesis* for Paper VII, not a result of the present paper: in this paper the spinor and scalar sectors are independent (Sec. 3), and the multiplicative form is therefore an ansatz rather than a derived consequence.

Why a product form is plausible. The two factors throttle distinct substrate channels: the scalar reserve R/R_0 limits the local bandwidth available to the substrate (Paper II), while the kinematic factor m/γ suppresses the comoving-frame phase rate of a moving packet (this paper). If, in the coupled rule of Paper VII, the scalar reserve enters the spinor evolution only as a multiplicative prefactor on the matter-hopping update A (so that $U_A \to U_A^{R/R_0}$ in some smooth sense) without modifying the gauge-twist update B that fixes the gap mass, then Eq. (21) follows directly. Other forms of coupling — e.g., an additive shift of the gap mass $m \to m + f(R)$, or a position-dependent c_{eff} — would yield qualitatively different combined factors.

The product form (21) agrees to leading order in both β/r and v^2 with the weak-field general-relativistic factor $\sqrt{1 - 2GM/c^2r - v^2/c^2}$, but differs at second order; if Paper VII's coupling produces the multiplicative ansatz, this is a structurally distinguishing prediction. We do *not*

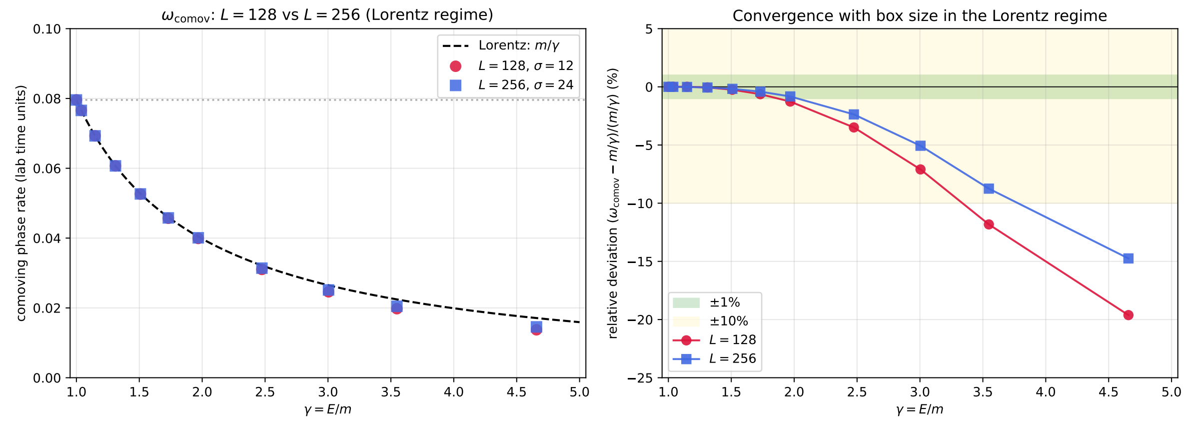

Figure 5: Box-size convergence of the operational comoving clock rate within the Lorentz regime $\gamma \leq 5$. *Left:* ω_{comov} vs γ for $L = 128, \sigma = 12$ (red circles) and $L = 256, \sigma = 24$ (blue squares), with the Lorentz prediction m/γ (dashed) and the rest-packet limit $m = 1/(4\pi)$ (grey dotted). *Right:* relative deviation $(\omega_{\mathrm{comov}} - m/\gamma)/(m/\gamma)$ for the two runs. The two box sizes agree to better than 0.5% in the strict Lorentz regime $\gamma < 2$; the $L = 256$ run shows smaller deviations at intermediate γ, consistent with the lattice corrections of Sec. 7 being a finite-resolution effect that shrinks with L. Data: `data/10_band_resolved_clock.json`.

verify Eq. (21) numerically here, and we do not present it as a prediction of the present paper. The verification is deferred to Paper VII (coupling) and Paper IV (gravitational phenomenology beyond g_{00}).

Calibration ledger (per SERIES_FEEDBACK §0.ter). This paper extends Papers I–II. New quantities and statuses:

Quantity	Status	Calibration source / future resolution
spinor $\psi_i \in \mathbb{C}^2$ on nodes	substrate field	defined; constrained by (F1)–(F4)
diagonal twist on links	substrate field	defined; constrained by (F-twist)
Floquet two-step $U_F = U_B U_A$	rule structure	forced by (F1)–(F4) at the minimal Pauli/Floquet representative
$\lambda = 1/(2\pi)$ (twist normalisation)	hypothesis (F-twist)	topological-minimality argument; microscopic origin open
$T = \tau_A + \tau_B$ (Floquet period)	calibrated	sets the analogue's tick scale; identified with empirical c via Planck anchoring
$m = \lambda/T$ (gap mass)	derived	from (F1)–(F4)+(F-twist) at gap point $(\pi, 0)$
$c_{\mathrm{eff}} = 1/T$ (ceiling)	derived	from finite per-tick bandwidth; identified with CODATA c via calibration of T
$\omega_{\mathrm{comov}} = m/\gamma$	derived	from Lorentz dispersion + Galilean shift in continuum limit

Rule loci touched and no-epicycle statement (per SERIES_FEEDBACK §0.quater). This paper extends Papers I–II at the following loci:

- *Nodes:* extended by a per-node spinor amplitude $\psi_i = (a_i, b_i)^\top \in \mathbb{C}^2$ (Sec. 3.4), in addition to the scalar reserve R_i of Paper I.
- *Links:* extended by a diagonal twist phase encoding the gauge holonomy of the (F-twist) sector.
- *Update rule:* extended by the Floquet two-step $U_F = U_B \cdot U_A$ of Eqs. (5)–(8), which alternates a matter-hopping update A and a gauge-twist update B. The Paper I rate-limiter and bipartite alternation are inherited as the scalar-sector update; the spinor sector is decoupled in this paper.
- *Substrate:* extended by a bipartite sublattice structure forced by (F3) (the two spinor components encode the two sublattices).
- *Configuration:* extended by Floquet-eigenstate wavepacket initialisation (Sec. 6) for the operational kinematic clock measurement.

No-epicycle statement. The scalar and spinor sectors are independent in this paper: the spinor rule does *not* relax or modify the principles (L1)–(L6) of Paper I or the hypotheses (H1)/(H2) of Paper II. No additional fields beyond those listed are introduced; no free parameters beyond the calibration ledger appear; the (F-twist) hypothesis is the only structural addition closing the construction. Future coupling between sectors is deferred to Paper VII as an explicitly labelled extension at the update locus, and the combined gravitational–kinematic factor of Sec. 8 is flagged as a *conjecture* rather than a silent assumption.

9 Limitations and scope

We do not derive the Lorentz group, special relativity, or any covariance result. We derive a kinematic time-dilation factor m/γ as a continuum limit of a specific lattice rule, near a band gap, conditional on the Floquet two-step structure adopted in Sec. 3.

We do not derive the topological twist normalisation (F-twist) of Sec. 3.1 from deeper substrate microphysics; this hypothesis fixes $\lambda = 1/(2\pi)$ and is the substantive postulate that closes the construction. What Sec. 3.1 establishes is narrower: within the class of local stateful-link substrates satisfying sub-tick causal locality, per-sub-tick unitarity, minimal complex amplitude carrying both orientation and phase, and first-order reversible propagation at gap closures, the rule structure of Eqs. (5)–(8) is selected as the minimal two-component Pauli/Floquet representative. Other representatives within the class exist (related by local unitary changes of basis, additional bands, or different UV corrections) but are low-energy-equivalent. Thus the remaining open assumption is not the spinor sector itself but the microscopic origin of unit flux quantisation in the diagonal-twist sector. Wilson [3] and Kogut–Susskind [4] discretisations are not the representatives adopted here; they implement the continuum Dirac structure through different lattice mechanisms and lie outside the minimal stateful-link Floquet class, but are useful comparators rather than violators of the axiomatic framework.

We do not derive Lorentz invariance *globally*. The dispersion is Lorentz-like only near the gap point $(\pi, 0)$. Other gap points (none in this 2D model, but several in higher dimensions) would carry their own effective masses and velocities. The substrate is not Lorentz-invariant as a whole; it is Lorentz-invariant in the low-energy effective theory at each gap.

2D scope of the present validation. This paper validates the 2D minimal kinematic prototype of the (F1)–(F4) + (F-twist) construction. The 3D extension and the gauge-sector content of Paper VI are *conjectured* on the basis of analogy with Weyl semimetals [6, 7] but *not established here*: the present numerical verification is on $L = 128$ in two spatial dimensions, with one gap point and one mass scale. Translation of the construction to three spatial dimensions, the emergence of Weyl points where the gap closes, and the gauge sector of Paper VI are natural extensions under investigation in forthcoming papers. Statements made in Sec. 3.9 about the universality class of the 3D rule should be read as conjectural until verified by explicit construction; we do not claim that (F1)–(F4) alone prove the existence of Weyl points in three dimensions.

We do not address the photon-like massless sector. The Floquet rule has no gapless mode in this 2D version; gapless propagating modes require a different gap structure or a different lattice topology, and are deferred to Paper VI (gauge sector). The gravitational-propagation consequences of photon-like modes (deflection, Shapiro delay, spatial metric) are deferred to Paper IV.

We do not address coupling between the spinor and scalar sectors. The scalar reserve R is uniform throughout the spinor's evolution in this paper. The combined gravitational-kinematic factor of Eq. (21) is conjectural pending Paper VII.

We do not derive Newton's G, the speed of light c, or the mass scale m in physical units. The lattice quantities are $\lambda = 1/(2\pi)$, $T = 2$ (in lattice units), giving $m = 1/(4\pi)$ and $c_{\text{eff}} = 1/2$ (in inverse lattice units). Conversion to physical units requires the same external Planck-scale anchoring as in Paper II's *Structural form of Newton's constant* section.

10 Open questions

Origin of the topological twist normalisation (F-twist) — the principal open problem. Sec. 3.1 derives the rule structure of Eqs. (5)–(8) from axioms (F1)–(F4), but does not derive (F-twist) — the unit-flux quantisation of the diagonal twist sector that fixes $\lambda = 1/(2\pi)$ — from deeper substrate microphysics. (F-twist) encodes the smallest non-trivial U(1) holonomy structure compatible with the bipartite sublattice forced by (F3), but the choice of unit flux quantum is asserted on topological-minimality grounds rather than derived.

What "topologically smallest non-trivial U(1) holonomy" means. On a bipartite lattice with the diagonal-twist sector forced by axiom (F3), the diagonal phase winding per unit area is quantised in integer multiples of a basic flux quantum (a standard consequence of single-valuedness of the gauge phase on closed plaquette loops, in the spirit of [8]). The choice $\lambda = 1/(2\pi)$ corresponds to one such unit; integer multiples $\lambda_n = n/(2\pi)$ with $n \geq 2$ would embed multiple flux quanta per plaquette diagonal, producing a non-minimal twist sector and additional UV scales without benefit at the low-energy effective theory. Fractional values $\lambda < 1/(2\pi)$ would correspond to non-integer winding classes, which are excluded by the same single-valuedness argument. "Topologically smallest non-trivial" is therefore shorthand for the minimal positive integer winding class compatible with the bipartite structure. What remains genuinely unexplained is *why this minimal class is the one realised* by the substrate, rather than $n = 0$ (no diagonal twist, no gap mass) or $n \geq 2$.

Conceivable routes towards a microscopic justification. Several routes are conceivable: (i) a topological invariant of the underlying lattice graph (e.g. a first Chern class associated with the diagonal-twist bundle, in the sense of [8]); (ii) a closure condition on a twisted-sector partition function over the substrate; (iii) a connection to Chern–Simons theory at the level of the effective long-distance description [9]; or (iv) a renormalisation-group fixed-point argument selecting unit flux as the unique stable normalisation. We do not pursue any of these here; we state the problem and emphasise that, because the gap mass $m = \lambda/T$ enters γ at every velocity, the kinematic result of this paper is conditional on the resolution of (F-twist). This is the principal open problem of the construction.

Coupling to the scalar sector. In a static gravitational background $R(r) < R_0$, does the spinor's effective mass m depend on R? Does the effective c_{eff}? The natural ansatz is $m_{\text{eff}}(r) = m\,(R/R_0)$ or similar, but this is not derived. Paper VII addresses this.

Lattice corrections at high γ. The functional form (20) is the leading quartic correction. Higher-order corrections, and the question of whether the deviation continues to be monotonically negative at all γ or saturates, are open. A scan to $\gamma \sim 30$ on $L = 512$ would settle the issue numerically.

Massless modes. In the present 2D model the spinor sector has no gapless mode; all bands are gapped. A 3D extension of the Floquet rule, as explored in the Weyl-semimetal analogy [7], exhibits Weyl points where the gap closes. These are the candidates for photon-like massless excitations and are the subject of Paper VI.

11 Code and data availability

The Python scripts in `code/` implement the Floquet two-step rule of Eq. (5) on a 2D periodic lattice. They are pure-numpy and depend on no external simulation framework.

- `01_wavepacket_propagation.py`: verifies that wavepackets propagate at the predicted group velocity.
- `02_floquet_dispersion.py`: computes the Floquet quasi-energy spectrum and verifies the gap structure.
- `04_kspace_projection.py`: tracks the lab-frame energy of a wavepacket via $\langle k_0|\psi(t)\rangle$.
- `05_comoving_clock_rate.py`: implements the Galilean-shifted dealiased projection of Sec. 5 and produces Table 3.

The numerical data are stored in `data/`. The repository is available at https://github.com/stanislasdewavrin/DDD-Paper-3-Inertia.

12 Outlook

Closing reaffirmation (analogue framing). This paper does not derive special relativity or Lorentz invariance. Instead, it shows that the *form* of the kinematic time-dilation factor can arise operationally from a discrete Floquet spinor rule on a lattice in the continuum limit near a band gap, conditional on the axioms (F1)–(F4) and the topological hypothesis (F-twist). The result is internal to the analogue.

This paper extracts an analogue of the kinematic time-dilation factor of special relativity from a lattice rule near a band gap, as the operational analogue of the bandwidth-clock identification of Paper II. The result and the rule are conditional on the axiomatic propagation framework (Sec. 3.1): axioms (F1)–(F4) uniquely select the two-step Floquet spinor structure, and the hypothesis (F-twist) fixes $\lambda = 1/(2\pi)$. The principal open task of the present paper is to derive (F-twist) from deeper substrate microphysics; until that is done, the gap mass $m = \lambda/T = 1/(4\pi)$ should be read as a consequence of the four propagation axioms together with the topological twist-normalisation hypothesis.

1. Paper IV — predictions and tests of the gravitational sector beyond the static spherically symmetric scenario: photon deflection, spatial metric, compact-source phenomenology. *Mixed.*
2. Paper V — gravitational predictions and falsifiers: where DDD differs from Newton/GR and how to test it. *Mixed.*

3. Paper VI — the gauge sector and emergent electromagnetism, in particular the gapless modes of the Floquet rule in 3D and the connection to Maxwell. *Speculative; coefficients uncalibrated.*

4. Paper VII — coupling between the scalar and spinor sectors. Combined static-gravity $\times$ kinematic factor; test of the multiplicative-product *conjecture* Eq. (21), and exploration of alternative coupling forms. *Conditional on Paper VI's gauge formulation.*

The result of this paper, taken on its own, is one structural fact: *the present paper does not derive Lorentz invariance.* It derives a Lorentz-form operational clock rate for wavepackets in the low-energy sector of a local unitary Floquet lattice rule. Specifically, a discrete lattice rule selected as the minimal two-component Pauli/Floquet representative of the (F1)–(F4) class produces, in the continuum limit near a band gap, the operational comoving frequency $\omega_{\text{comov}} = m/\gamma$ at moving excitations. The interpretation is intrinsic to the rule; no Lorentz invariance, no metric, and no Lorentzian proper-time formula is assumed; what is identified is the operational substrate quantity, not a metric postulate. The factor that emerges has the same form as in special relativity, accompanied by lattice corrections at high γ that are calculable and, in principle, falsifiable.

References

[1] S. Dewavrin. Discrete drainage dynamics i: Foundations of the scalar reserve substrate. In preparation; preprint available from the author, 2026.

[2] S. Dewavrin. Discrete drainage dynamics ii: Bandwidth identification and gravitational time dilation. In preparation; preprint available from the author, 2026.

[3] K. G. Wilson. Confinement of quarks. *Phys. Rev. D*, 10:2445, 1974.

[4] J. Kogut and L. Susskind. Hamiltonian formulation of Wilson's lattice gauge theories. *Phys. Rev. D*, 11:395, 1975.

[5] A. Eckardt. Colloquium: Atomic quantum gases in periodically driven optical lattices. *Rev. Mod. Phys.*, 89:011004, 2017.

[6] X. Wan, A. M. Turner, A. Vishwanath, and S. Y. Savrasov. Topological semimetal and Fermi-arc surface states in the electronic structure of pyrochlore iridates. *Phys. Rev. B*, 83:205101, 2011.

[7] N. P. Armitage, E. J. Mele, and A. Vishwanath. Weyl and Dirac semimetals in three-dimensional solids. *Rev. Mod. Phys.*, 90:015001, 2018.

[8] D. J. Thouless, M. Kohmoto, M. P. Nightingale, and M. den Nijs. Quantized Hall conductance in a two-dimensional periodic potential. *Phys. Rev. Lett.*, 49:405, 1982.

[9] E. Witten. Quantum field theory and the jones polynomial. *Commun. Math. Phys.*, 121:351, 1989.